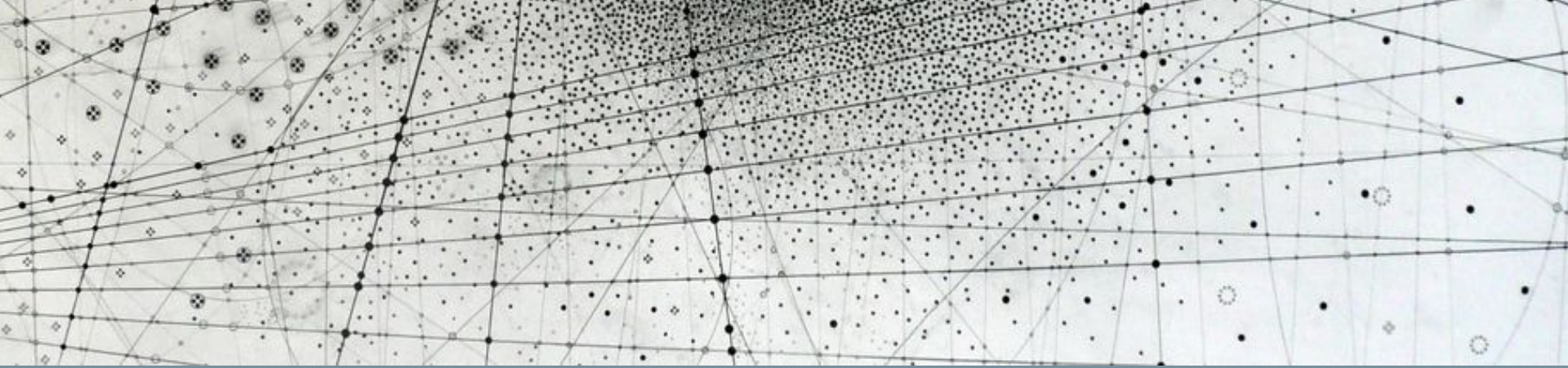




Research Methods in Machine Learning & Physics

Quick recap

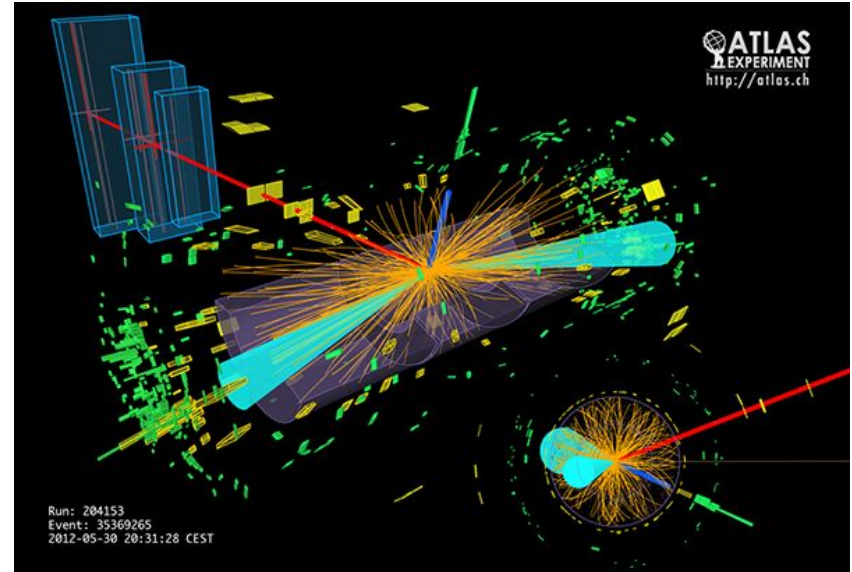
Simulation-based Inference

"Given my data x , what is the most favored value of θ ?"

Theory (θ)

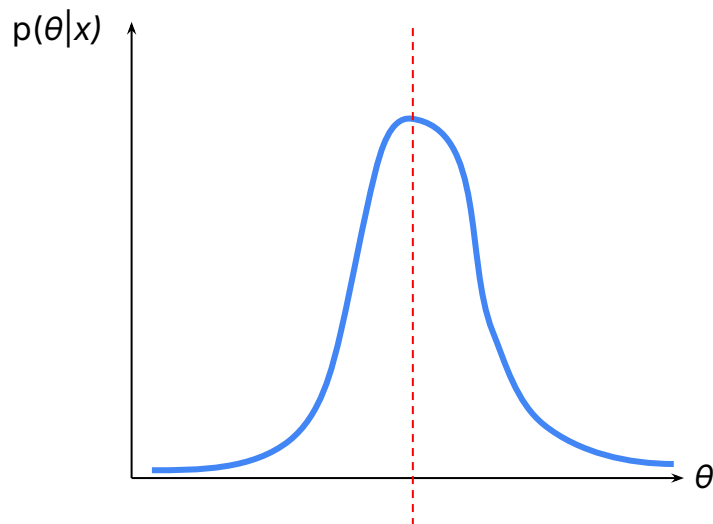
$$\begin{aligned}\mathcal{L} = & -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ & + i \bar{\Psi} \not{D} \Psi + h.c. \\ & + \bar{\Psi}_i y_{ij} \Psi_j \phi + h.c. \\ & + |\nabla_\mu \phi|^2 - V(\phi)\end{aligned}$$

Data (x)



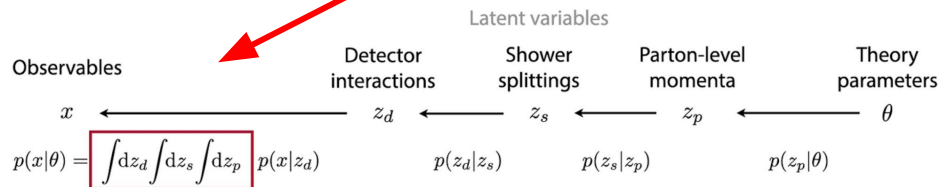
Simulation-based Inference

"Given my data x , what is the most favored value of θ ?"



Higgs mass = 125 GeV

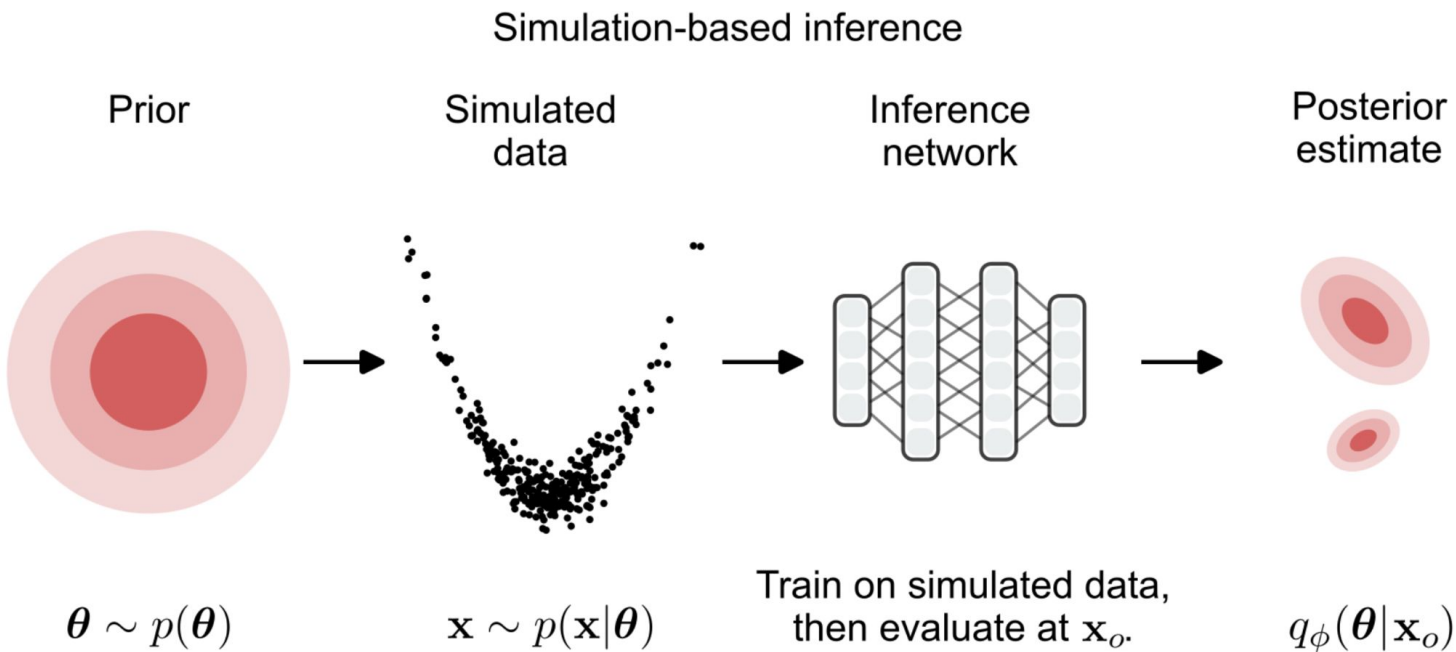
$$p(\theta|x) \propto \underline{p(x|\theta)} p(\theta)$$



It's infeasible to calculate the integral over this enormous space!

Simulation-based Inference

Using neural networks to facilitate statistical inference on simulated data:



Simulation-based Inference

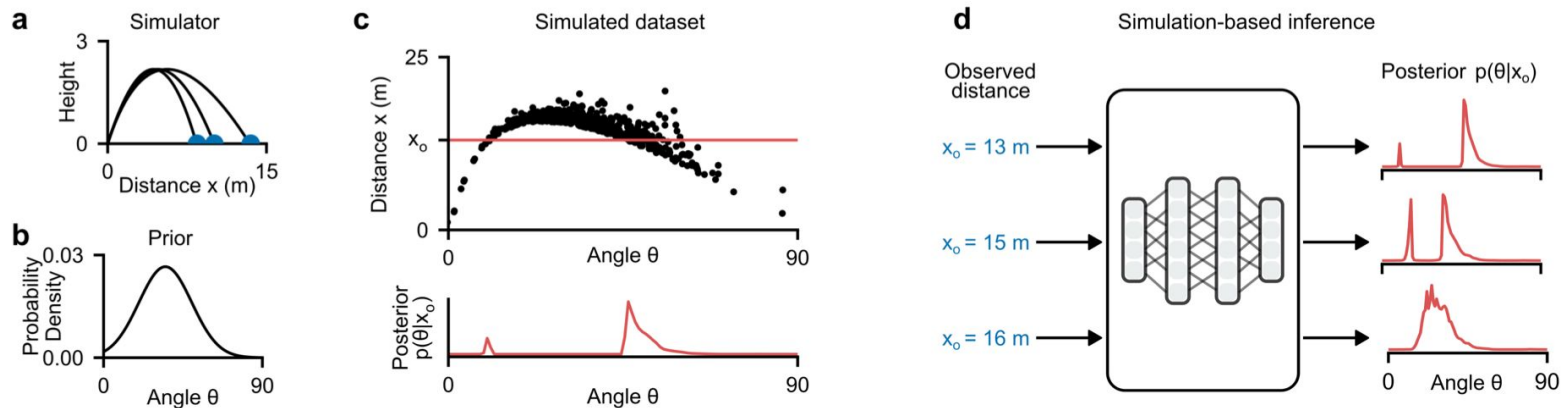


Figure 2. Introductory example: SBI for a ball throwing simulator. (a) Trajectories of the ball-throwing simulator. Only the final distance (blue) is used for inference. (b) Prior distribution for the throwing angle θ . (c) Top: Simulated dataset of N throws, showing pairs of (θ, x) samples from the joint distribution $p(\theta, x)$ and a hypothetical observed distance $x_o = 13$ m (red line). Bottom: Posterior distribution $p(\theta|x_o)$, i.e., the conditional distribution of parameters θ given the observed distance x_o , visualized as a horizontal “cut” through the joint distribution. Note the bimodal shape, reflecting that $x_o = 13$ m could be produced both by low and high (but not by intermediate) angles. (d) Amortized SBI. An inference network is trained on simulated data to learn the mapping from an input distance x to the corresponding posterior distribution $p(\theta|x)$. Once it is trained, it can return the posterior for new observed distances x_o without needing further simulations or retraining. Examples for three values of x_o are shown.

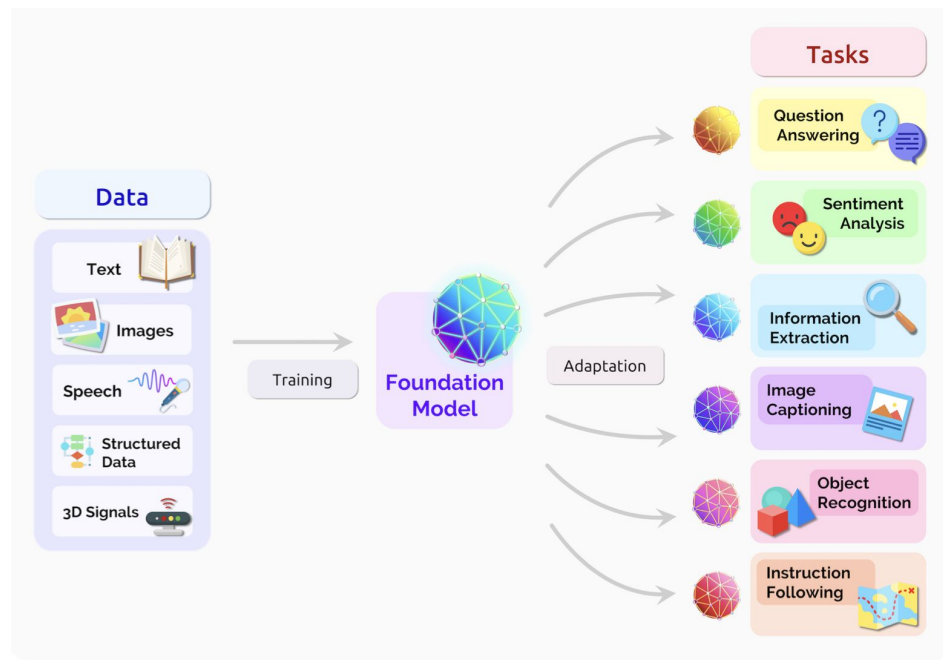
Foundation Models, Part I

Foundation models are “models trained on broad data (generally using self-supervision at scale) that can be adapted to a wide range of downstream tasks.”

On the Opportunities and Risks of Foundation Models

Rishi Bommasani* Drew A. Hudson Ehsan Adeli Russ Altman Simran Arora
Sydney von Arx Michael S. Bernstein Jeannette Bohg Antoine Bosselut Emma Brunskill
Erik Brynjolfsson Shyamal Buch Dallas Card Rodrigo Castellon Niladri Chatterji
Annie Chen Kathleen Creel Jared Quincy Davis Dorottya Demszky Chris Donahue
Moussa Doumbouya Esin Durmus Stefano Ermon John Etchemendy Kawin Ethayarajh
Li Fei-Fei Chelsea Finn Trevor Gale Lauren Gillespie Karan Goel Noah Goodman
Shelby Grossman Neel Guha Tatsunori Hashimoto Peter Henderson John Hewitt
Daniel E. Ho Jenny Hong Kyle Hsu Jing Huang Thomas Icard Saahil Jain
Dan Jurafsky Pratyusha Kalluri Siddharth Karamcheti Geoff Keeling Fereshte Khani
Omar Khattab Pang Wei Koh Mark Krass Ranjay Krishna Rohith Kuditipudi
Ananya Kumar Faisal Ladhak Mina Lee Tony Lee Jure Leskovec Isabelle Levent
Xiang Lisa Li Xuechen Li Tengyu Ma Ali Malik Christopher D. Manning
Suvar Mirchandani Eric Mitchell Zanele Munyikwa Suraj Nair Avaniika Narayan
Deepak Narayanan Ben Newman Allen Nie Juan Carlos Niebles Hamed Nilforoshan
Julian Nyarko Gray Ogut Laurel Orr Isabel Papadimitriou Joon Sung Park Chris Piech
Eva Portelance Christopher Potts Aditi Raghunathan Rob Reich Hongyu Ren
Frieda Rong Yusuf Roohani Camilo Ruiz Jack Ryan Christopher Ré Dorsa Sadigh
Shiori Sagawa Keshav Santhanam Andy Shih Krishnan Srinivasan Alex Tamkin
Rohan Taori Armin W. Thomas Florian Tramèr Rose E. Wang William Wang Bohan Wu
Jiajun Wu Yuhuai Wu Sang Michael Xie Michihiro Yasunaga Jiaxuan You Matei Zaharia
Michael Zhang Tianyi Zhang Xikun Zhang Yuhui Zhang Lucia Zheng Kaitlyn Zhou
Percy Liang*¹

Center for Research on Foundation Models (CRFM)
Stanford Institute for Human-Centered Artificial Intelligence (HAI)
Stanford University



A slightly more relaxed definition of foundation models in science:

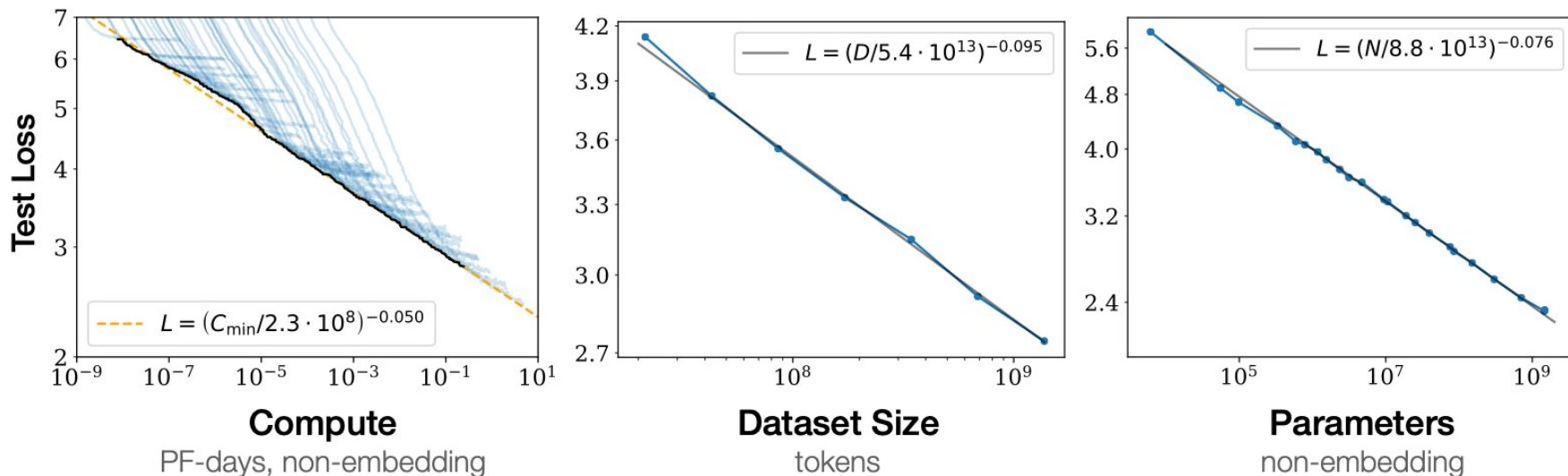
1. It has been pretrained on a large amount of data, with the **explicit intention of creating a rich latent representation** that serves as a foundation for other tasks;
2. It uses this latent representation to **finetune the model to a different downstream task** (be it the same type of task on a different dataset, or a different task altogether);
3. The **final performance on the downstream task improves** when the latent representation is used, compared to if the model is initialized without this representation.

Examples of foundation models in **industry**

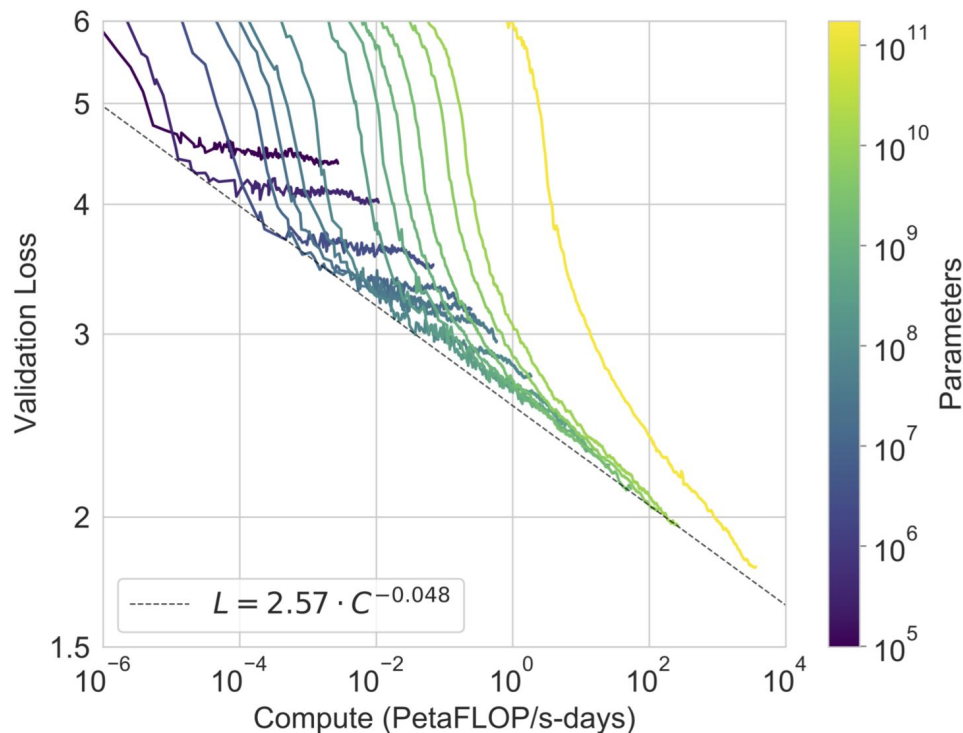
- BERT
- GPT
- CLIP
- DALL-E
- SORA
- Stable Diffusion
- Gemini
- Claude
- DeepSeek
- ...

Scaling laws

Large-scale foundation models benefit from seemingly unstoppable power laws of scale:



Scaling laws



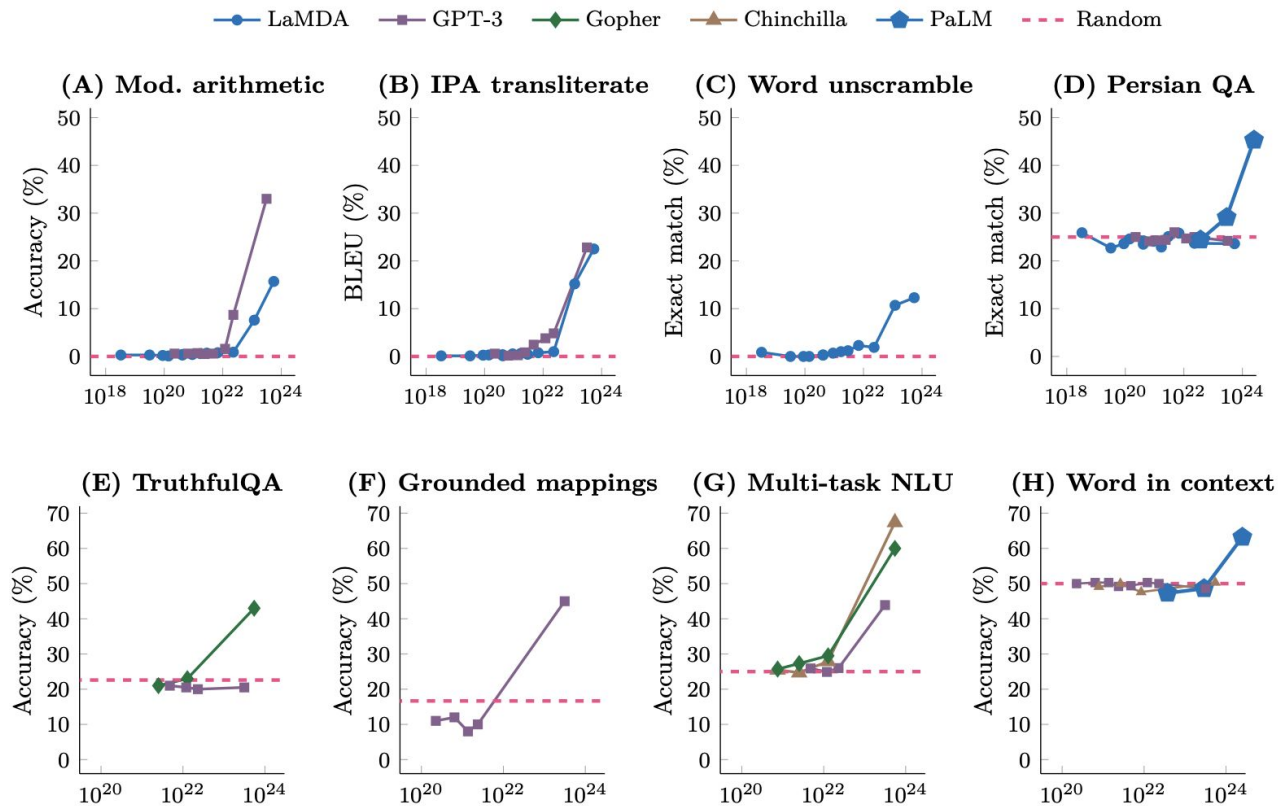
The “Death” of Scaling Laws

Scaling laws have recently become a popular (and contentious) topic within AI research. As we have seen throughout this overview, scaling has fueled most improvements in AI throughout the age of pretraining. As the pace of model releases and improvements slowed in the latter half of 2024¹¹, however, we began to see widespread questioning of model scaling, seeming to indicate that AI research—and scaling laws in particular—could be hitting a wall.

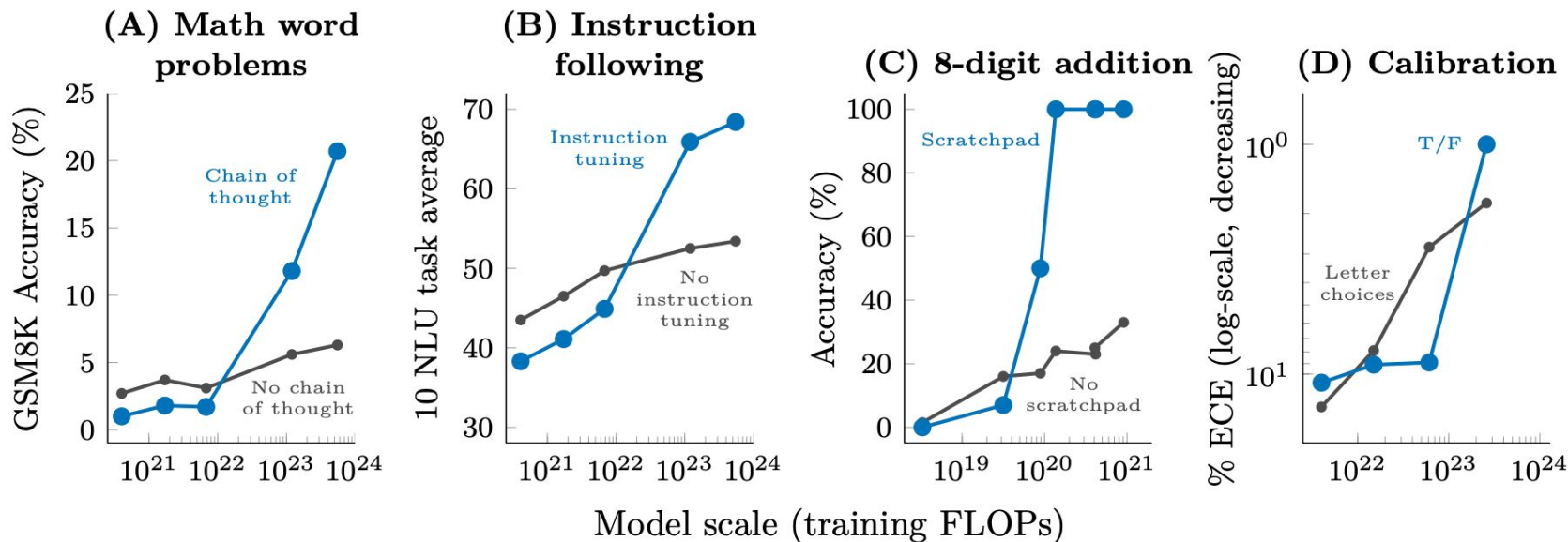
- [Reuters](#) claims that OpenAI is shifting their product strategy due to hitting a plateau in scaling current methods.
- [The Information](#) claims that the rate of improvement in GPT models is starting to slow down.
- [Bloomberg](#) highlights the difficulties being faced by several frontier labs in trying to build AI that is more capable.
- [TechCrunch](#) claims that scaling is starting to yield diminishing returns.
- [Time](#) publishes a nuanced essay that highlights various contributing factors to the narrative that AI research is slowing down.
- Ilya Sutskever states that “pretraining as we know it will end” during his [test of time award speech](#) at NeurIPS’24.

At the same time, many experts are arguing the opposite. For example, [Dario Amodei](#) (Anthropic CEO) has said that scaling is “probably... going to continue”, while Sam Altman has continued pushing the narrative that “[there is no wall](#)”.

Emergence of new skills at scale

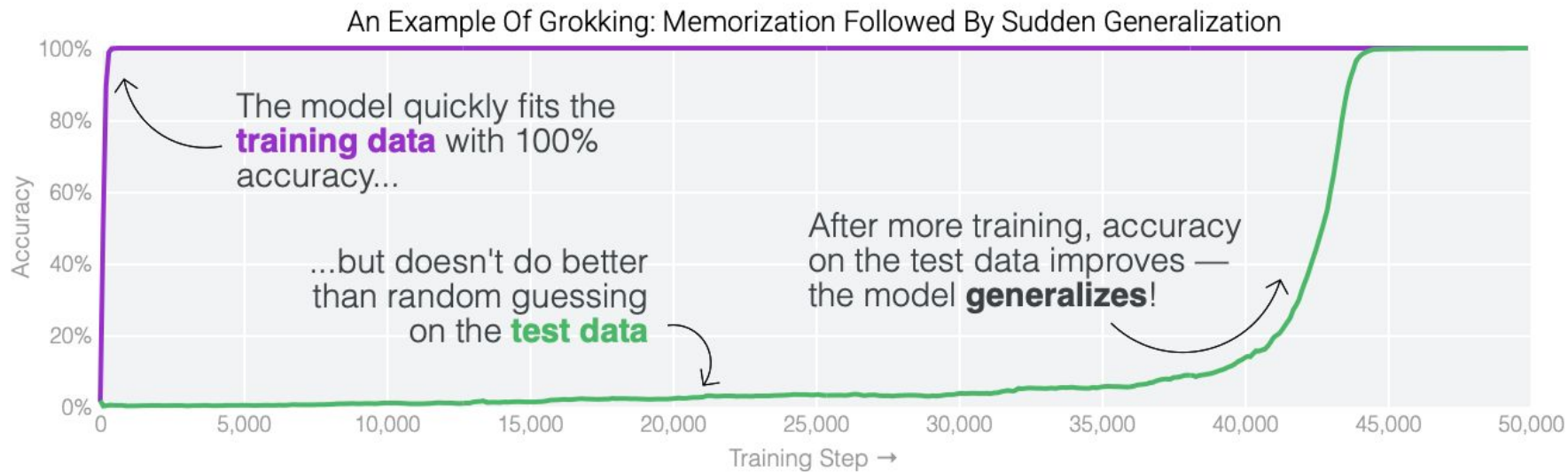


Emergent skills in chain-of-thought reasoning & more



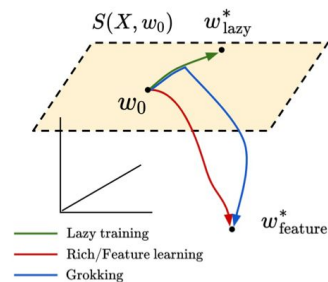
"Grokking"

Delayed generalization ability on test data after a period of overtraining:



GROKKING AS THE TRANSITION FROM LAZY TO RICH TRAINING DYNAMICS

Tanishq Kumar Blake Bordelon Samuel J. Gershman* Cengiz Pehlevan*
Harvard University



Fine-tuning

Fine-tuning typically involves “nudging” your model in a new direction after the hefty “pre-training” stage – you don’t want to modify it too much, but just enough to adapt to a new setting.

e.g. “freeze” the weights of several layers (so they don’t change), but leave a few layers “unfrozen” so that they can adapt to new high-level features.

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



In-context learning, i.e. learning without training!

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese =>                    ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer   ← example
3 cheese =>                    ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer   ← examples
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese =>                    ← prompt
```

In-Context Learning: Learning *without* weight updates

9 - 2 = 11, 13 - 7 = 20, 2 - 5 = 7, 4 - 1 = ??

Input

Reference
text-davinci-003
gpt-3.5-turbo
gpt-4
Claude 2
Human

Kor kancing wa me an tur teba ma patin.

My ankle bone, I fell and wounded it.
Even if I fall, my ankle bone is wounded.
I dream of a watch falling soon and injuring.
This ankle of mine is falling and progressively getting wounded.
This ankle of mine fell down while I was walking, and it got wounded.
This ankle, I injured by falling.

Figure 4: **A qualitative example of kgv→eng translation.** Outputs are from the highest-context setting for each model. text-davinci-003 translates the vocabulary correctly but hallucinates an “even if” structure. gpt-3.5-turbo is distracted by irrelevant vocabulary. gpt-4 shows an interesting error where it interprets the definition of *teba*, a progressive aspect marker, into the word “progressively”. Claude 2 is phrased strangely yet is almost correct, but the input does not mention walking. The human translation is correct but uses a slightly unnatural phrasing to reflect topicalization and progressive aspect in the input.

In-context learning, i.e. learning without training!

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

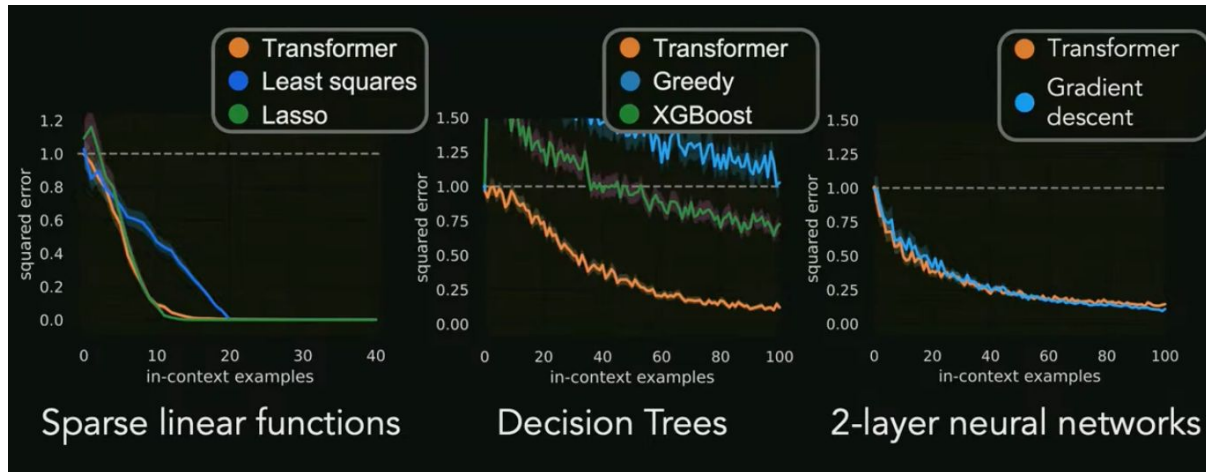
In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese => ..... ← prompt
```



How is this possible?? → approximating gradient descent?

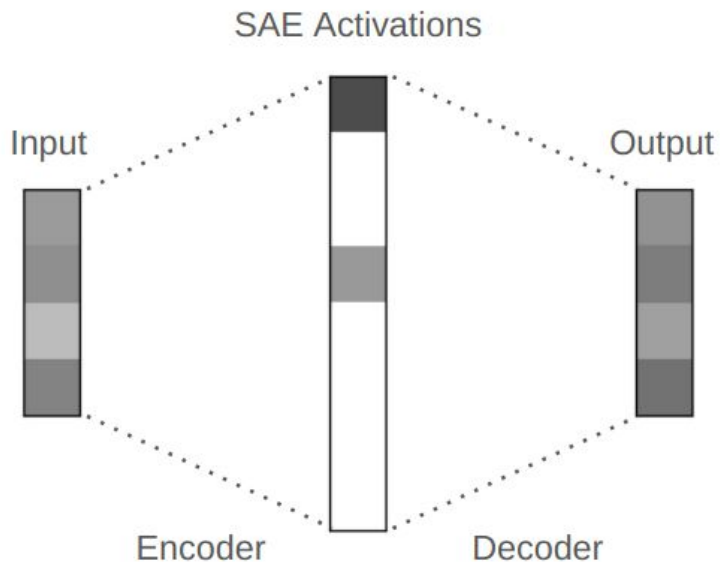
Transformers Learn In-Context by Gradient Descent

Why Can GPT Learn In-Context?
Language Models Implicitly Perform Gradient Descent as
Meta-Optimizers

2. Linear self-attention *can* emulate gradient descent on a linear regression task

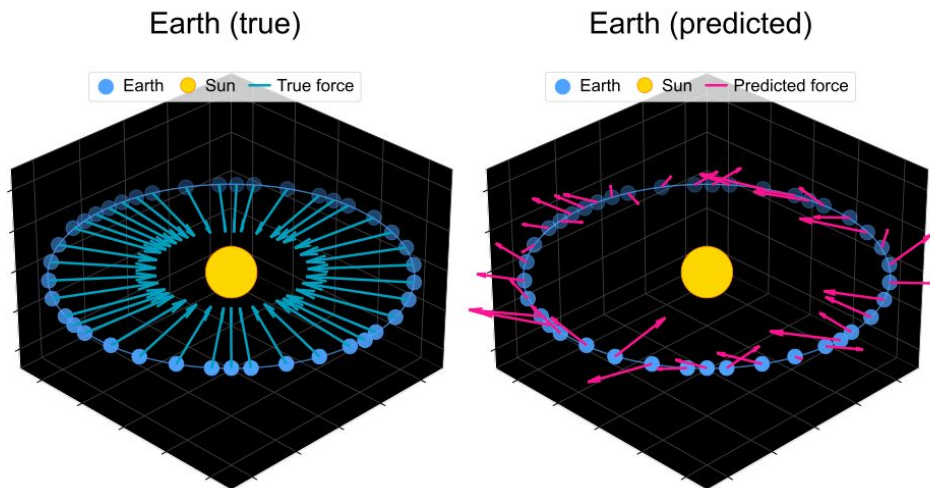
Interpretability

Sparse Autoencoders (SAEs) can reveal important features



What Has a Foundation Model Found? Using Inductive Bias to Probe for World Models

Keyon Vafa¹ Peter G. Chang² Ashesh Rambachan² Sendhil Mullainathan²



In-class work

1. Design some experiments to measure in-context learning performance on an LLM of your choice (think about what tasks *aren't* in the training data!)
2. Give some examples of learned semantic features from Anthropic's recent work in sparse autoencoders and how they found them.
3. Explain how AstroCLIP fine-tunes its embedding for one or two downstream tasks.

We'll reconvene at the top of the hour to share our highlights.